

LOXOPROFEN SODIUM HYDRATE

Containing Loxoprofen Sodium Hydrate (JP) 113.4mg (100mg as Anhydride)



LOXOTOP PAP 100mg Patch

FOR EXTERNAL USE ONLY

QUALITATIVE AND QUANTITATIVE COMPOSITION

QUALITATIVE COMPOSITION

Generic Name

Loxoprofen Sodium Hydrate (JAN) / Loxoprofen (INN)

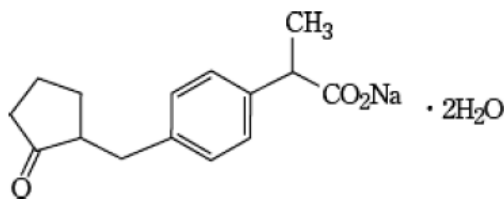
Chemical Name

Monosodium 2- { 4 - [(2-oxocyclopentyl) methyl]phenyl} propanoate dihydrate

Molecular Formula $C_{15}H_{17}NaO_3 \cdot 2H_2O$

Molecular Weight 304.31

Structural Formula



Description

Loxoprofen sodium hydrate occurs as a white to yellowish white, crystal or crystalline powder. It is very soluble in water and in methanol, freely soluble in ethanol (95), and practically insoluble in diethyl ether. A solution of loxoprofen sodium hydrate (1 in 20) does not show optical rotation.

QUANTITATIVE COMPOSITION

LOXOTOP PAP 100mg Patch

A patch containing 113.4mg (100mg as anhydride) of loxoprofen sodium hydrate in 10g of ointment (a sheet of 140 cm²).

PHARMACEUTICAL FORM

LOXOTOP PAP 100mg Patch

Patch preparation comprising ointment, uniformly spread over a support with the surface of the ointment covered by a liner. The ointment surface is white to pale yellow in color. The ointment has a fragrance of mentha oil.

CLINICAL PARTICULARS

1. Therapeutic indications

LOXOTOP PAP 100mg Patch is indicated for anti-inflammation and pain relief in the following diseases and symptoms: Osteoarthritis, myalgia, posttraumatic swelling and pain.

2. Posology and method of administration

LOXOTOP PAP 100mg Patch should be applied once daily to the affected area.

3. Contraindications

LOXOTOP PAP 100mg Patch is contraindicated in the following patients.

- Patients with a history of hypersensitivity to any ingredients of LOXOTOP PAP 100mg Patch
- Patients with or with a history of aspirin-induced asthma (induction of asthmatic attack with nonsteroidal anti-inflammatory-analgesics, etc.) [May induce an aspirin-induced asthmatic attack.]

4. Special warnings and precautions for use

WARNING

Risk of GI Ulceration, Bleeding and Perforation with NSAID

Serious GI toxicity such as bleeding, ulceration and perforation can occur at any time, with or without warning symptoms, in patients treated with NSAID therapy. Although minor upper GI problems (e.g. dyspepsia) are common, usually developing early in therapy, prescribers should remain alert for ulceration and bleeding in patients treated with NSAIDs even in the absence of previous GI tract symptoms.

Studies to date have not identified any subset of patients not at risk of developing peptic ulceration and bleeding. Patients with prior history of serious GI events and other risk factors associated with peptic ulcer disease (e.g. alcoholism, smoking, and corticosteroid therapy) are at increased risk. Elderly or debilitated patients seem to tolerate ulceration or bleeding less than other individuals and account for most spontaneous reports for fatal GI events.

4.1. Careful Administration

LOXOTOP PAP 100mg Patch should be administered with caution in the following patients: Patients with bronchial asthma (as the disease state may be exacerbated.)

4.2 Clinically Significant Adverse Reactions

Shock, anaphylaxis: Shock or anaphylaxis (decreased blood pressure, urticaria, laryngeal oedema, dyspnoea, etc.) may occur. Patients should be carefully monitored. If any abnormalities are observed, use of this drug should be discontinued immediately and appropriate measures should be taken.

4.3. Important Precautions

- It is important to note that treatment with anti-inflammatory- analgesic agents is a symptomatic treatment, not a causal treatment.
- For treatment of skin inflammation caused by infectious disease, LOXOTOP PAP 100mg Patch may cause a risk of masking the signs and symptoms of the infectious disease, therefore, the appropriate antibacterial and/or antifungal drugs must be administered in combination with LOXOTOP PAP 100mg Patch to the affected part of the skin, under careful observation and caution.
- Therapies other than drug treatment must also be considered in using LOXOTOP PAP 100mg Patch in the management of chronic diseases (osteoarthritis and others). The patient's clinical condition should be closely monitored with caution against the development of adverse reactions.

4.4. Use in the Elderly

LOXOTOP PAP 100mg Patch should be used with careful monitoring of dermal condition of application area, in the elderly patients.

4.5. Pediatric Use

The safety of LOXOTOP PAP 100mg Patch in low birth weight infants, newborn infants, infants and toddlers, children and adolescents has not been established (because there has been little experience of its use in pediatric patients).

4.6. Precautions Concerning Use

Site of application:

- Do not apply LOXOTOP PAP 100mg Patch onto area of damaged or non-intact skin or mucosal membrane.
- Do not apply LOXOTOP PAP 100mg Patch onto area of eczema and rash.

5. Interactions with other Medicinal Products and other Forms of Interactions

Not applicable to both contraindications and precautions for coadministration.

6. Pregnancy and Lactation

- LOXOTOP PAP 100mg Patch should be administered to women who are or are possibly pregnant only when the anticipated therapeutic benefits are considered to outweigh any potential risk. (The safety of LOXOTOP PAP 100mg Patch in these populations has not been established.)
- The following information is for reference only, the adverse reaction is not reported on the use of this product. It has been reported that constriction of the fetal ductus arteriosus occurred when woman in second and/or third trimester of pregnancy received cyclooxygenase inhibitors.
- Renal impairment and decreased urine output in foetuses as well as accompanying oligohydramnios have been reported following use of cyclooxygenase inhibitors (oral dosage form or suppository) in pregnant women.

7. Effects on Ability to Drive and Use Machines

There is no data available on effects on ability to drive and use machines.

8. Undesirable Effects

Clinically Significant Adverse Reactions

Shock, anaphylaxis (The incidence unknown)

Other adverse reactions

	0.5% to <3%	<0.5%	Incidence unknown
Hypersensitivity	Pruritus, erythema, contact dermatitis, rash	Haemorrhage subcutaneous, skin irritation, pigment precipitation	Blister, swelling
Gastrointestinal		Stomach discomfort, upper abdominal pain, diarrhoea, loose stools	
Hepatic		Increased AST(GOT), increased ALT(GPT), increased γ -GTP	

Other		Edema	
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9. Overdose

There is no data available on overdose.

PHARMACOLOGICAL PROPERTIES

1. Pharmacodynamic properties

1.1. Mechanism of Action

After being absorbed transdermally, loxoprofen sodium hydrate is biotransformed into an active metabolite *trans*-OH form (SRS coordination) to exert its anti-inflammatory and analgesic effects in acute inflammations, chronic inflammations and pain.

1.1.1. Anti-inflammatory effects

LOXOTOP PAP 100mg Patch has been demonstrated significant anti-inflammatory effects on both acute inflammations such as carrageenan-induced edema (rat) and chronic inflammations such as adjuvant arthritis (rat).

1.1.2. Analgesic effects

LOXOTOP PAP 100mg Patch has been demonstrated to show the analgesic effect in the Randall-Selitto test (inflamed paw pressuring method: rat), and in the chronic adjuvant arthritis pain test (rat).

2. Pharmacokinetic properties

2.1. Plasma Concentration

Each of 14 Japanese healthy male adult volunteers was subjected to repeated administration of two LOXOTOP PAP 100mg Patch on the back once a day for five days, showing detection of loxoprofen and its *trans*-OH form (active metabolite) in the plasma immediately after the start of the administration with their plasma concentrations gradually increasing over the period of the administration before reaching steady state in four or five days after the administration at low levels compared to those achieved by the equivalent oral dose of the same drug until, upon the discontinuation of the administration, they rapidly disappeared from the plasma to concentrations below the limit of quantitative determination (LQD).

2.2. Tissue Penetration

Administration of 3.5cm² of the LOXOTOP PAP 100mg Patch (containing ¹⁴C-loxoprofen) to the rat dorsal skin for a time period of 24 hours showed that the concentration of radioactivity in the skeletal muscle immediately under the patch administered area of the skin was 3.6- to 24-fold higher than that in the

skeletal muscle under the patch non-administered area of the skin, confirming the formation of the *trans*-OH form of ¹⁴C-loxoprofen as its active metabolite.

2.3. Metabolism

An in vitro test of loxoprofen sodium hydrate for metabolism inhibition using human hepatic microsome showed that the drug did not inhibit the metabolism of substrates for cytochrome P450 enzymes (CYP1A1/2, 2A6, 2B6, 2C8/9, 2C19, 2D6, 2E1 and 3A4) even at a concentration of 200µM equivalent to 1,000 times or more than the maximum plasma concentration observed by administration of LOXOTOP PAP 100mg Patch in a dose of one patch a day.

2.4. Urinary Excretion

Each of 14 Japanese healthy male adult volunteers was subjected to repeated administration of two LOXOTOP PAP 100mg Patch on the back once a day for five days, showing that the daily urinary excretion of loxoprofen, its *trans*-OH form (active metabolite) and *cis*-OH form remained almost constant after the elapse of 24 hours following the administration with the total cumulative excretion rate of 2.67% from the start of the administration until 48 hours after the discontinuation of the administration.

PHARMACEUTICAL PARTICULARS

1. List of excipients

LOXOTOP PAP 100mg Patch

Mentha oil, polysorbate 80, titanium oxide, tartaric acid, disodium edetate hydrate, concentrated glycerin, carmellose sodium, talc, dried aluminum hydroxide gel, crotamiton, partially neutralized polyacrylate, starch grafted acrylate 300, emulsion of methyl acrylate and 2-ethylhexyl acrylate copolymer

1.1. Shelf life

24 months

1.2. Special precautions for storage

LOXOTOP PAP 100mg Patch must be stored in light-proof tightly-sealed container. Do not store above 30°C.

1.3 Instructions for use/handling

1.3.1. Expiration date for use

LOXOTOP PAP 100mg Patch must be used before the expiration date indicated on the package or label.

1.3.2. Instructions for use/handling

After removing the patch from the pouch for use, be sure to close it securely by gently pressing both sides of its zip-lock portion against each other.

Pack Sizes: LOXOTOP PAP 100mg Patch is available in 10 sachets per box with 5 patches in each sachet.

MANUFACTURER

Lead Chemical Co., Ltd.

Hisagane Plant

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MARKETING AUTHORISATION NUMBER (MAL Number)

MAL20046063ARZ

DATE OF REVISION OF THE TEXT

23 May 2025

PRODUCT REGISTRATION HOLDER

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